



IILM University, Greater Noida
Internship Report Presentation
Thiranex C++ Programming

Virtual Internship

Harshit Pandey

Roll No. 25scs1003000929 | B.Tech CSE | Batch 2025-2029

19th August – 18th September 2026

Internship & Profile Overview

Role & Duration

Intern – C++ Programming

19 August 2026 – 18 September 2026

Work Structure

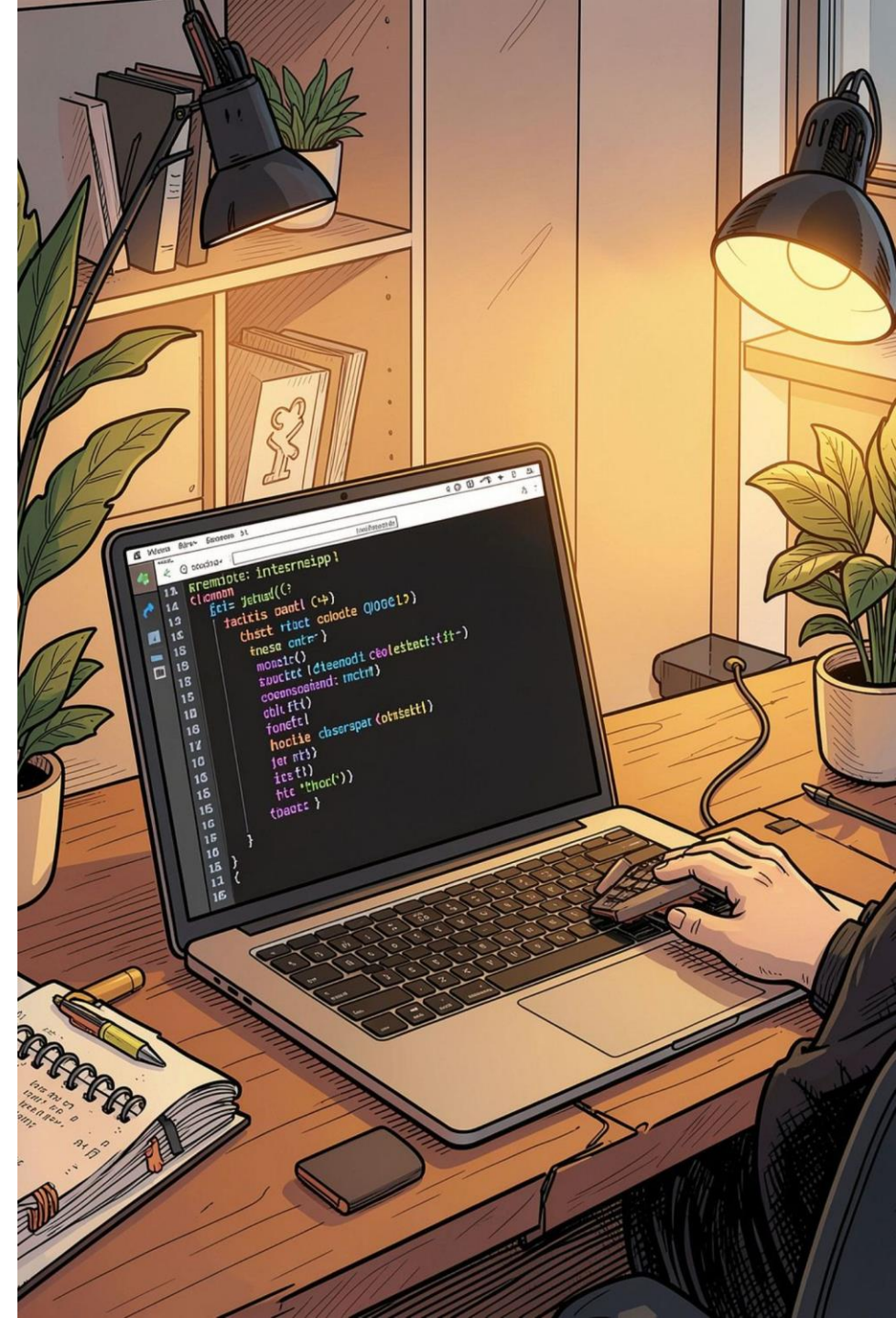
Remote, project-based work with structured milestones and industry mentorship.

Core Technical Focus

Hands-on application of C++ fundamentals, OOP, File Handling, and CLI logic.

Industry Preparation

Building production-ready, memory-safe console applications without framework dependencies.



Key Objectives & Problem Areas

The internship focused on three core technical challenges that mirror real-world C++ development constraints.



Persistent Storage

Implementing data persistence **without relational database engines** using raw C++ file streams (`fstream`).



OOP & Abstraction

Structuring modular programs using **classes, object abstraction, and encapsulation** for secure data access.



CLI Application UI

Designing **interactive, menu-driven terminal interfaces** with dynamic state rendering and input validation.

Development Methodology

01

Requirement Analysis

Analysing domain constraints, operational requirements, and defining object models.

02

Data Design

Constructing C++ structures and classes with private fields and clean public interfaces.

03

Storage Logic

Implementing append, read-sequential, and write override disk operations with `fstream`.

04

Verification & Debugging

Testing boundary cases, stream exceptions, and menu navigation loops.



C++ Technology Stack

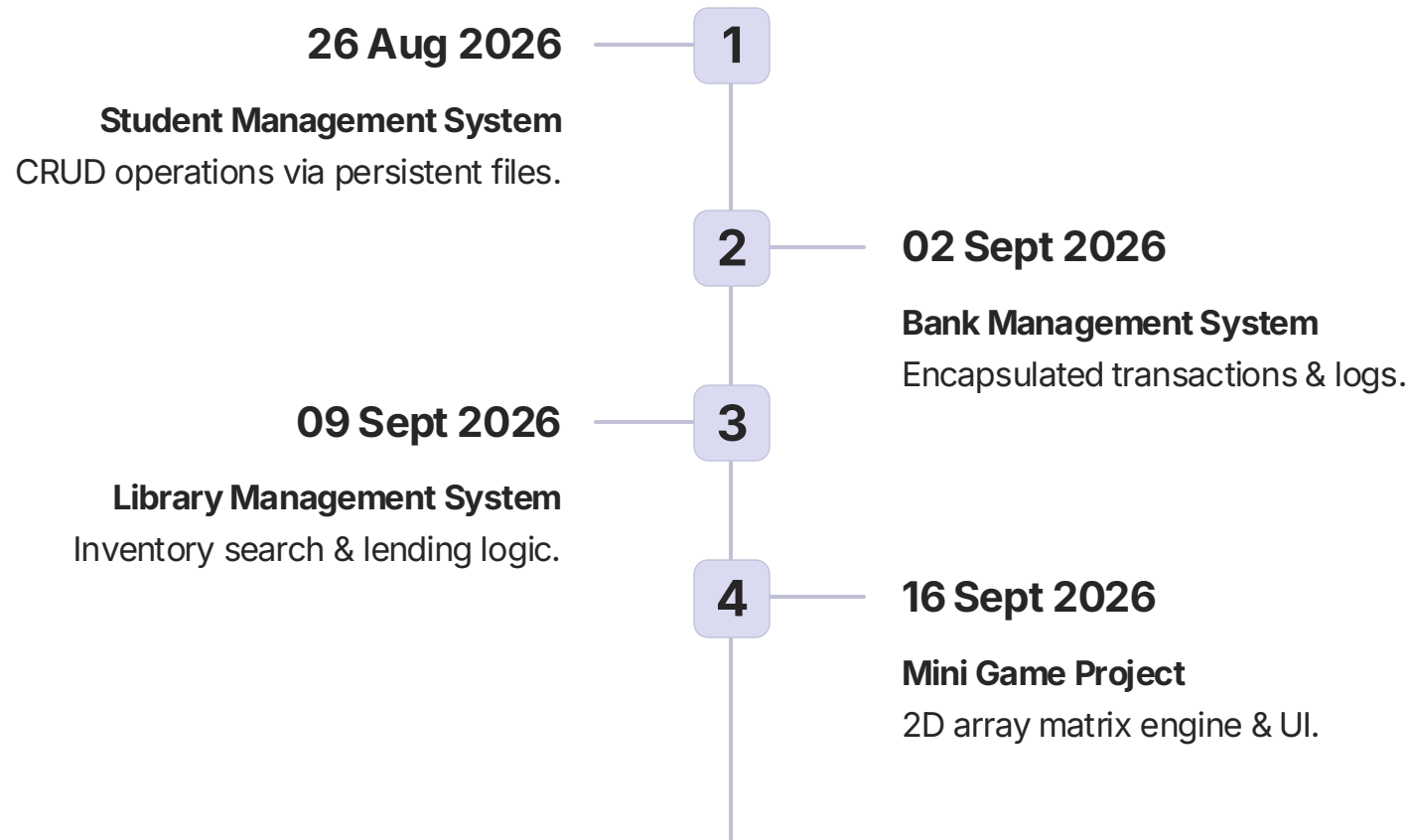
Core Stack & Tools

- **Language Standards:** C++11 / C++17
- **Development IDE:** VS Code / Code::Blocks / GCC Compiler
- **Version Control:** Git & GitHub
- **Standard Libraries:** iostream, fstream, vector, string, iomanip, algorithm



All projects were built using native C++ without external database or framework dependencies, ensuring a deep understanding of low-level memory management and file I/O operations.

Project Milestone Timeline



📋 All four milestones were completed and submitted on schedule, demonstrating consistent progress throughout the one-month internship.

Task 1 & Task 2 — Student & Bank Management Systems

Student Management System

Developed a console program performing **Create, Read, Update, and Delete** operations on student records. Integrated C++ file streams for persistent disk storage and tabular search lookup formatting.

Bank Management Application

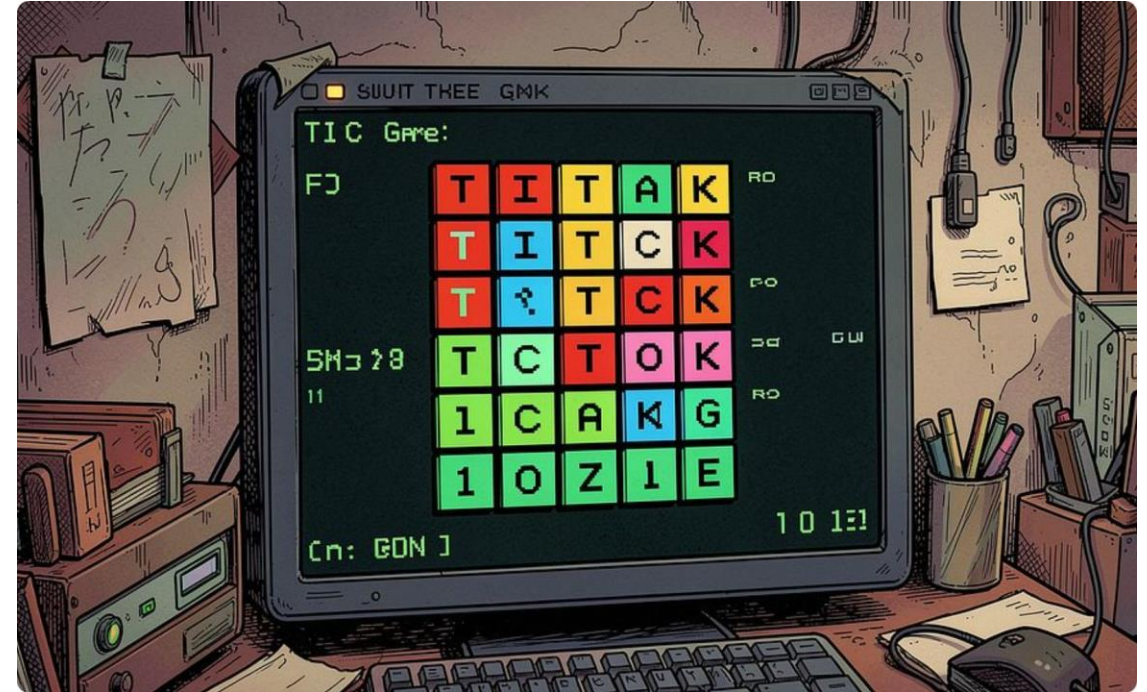
Simulated core banking operations — **Deposit, Withdrawal, and Balance Inquiry** — with class encapsulation. Enforced data validation rules and secure transaction record logs stored persistently in file storage.

Task 3 & Task 4 — Library System & Mini Game



Library Management System

Created a comprehensive inventory system modelling book items and member lending history. Implemented **fast search lookup algorithms** for title/author filtering and real-time issue/return updates.



Interactive Mini Game

Engineered a CLI game (Tic Tac Toe / Snake) using **2D matrix arrays and interactive dynamic loops**. Programmed real-time win/loss state verification, score tracking, and smooth board re-rendering cycles.

Performance Summary & Core Technical Learnings

100%

Milestone Submissions

All submissions on track before scheduled deadlines.

4

Core Software Modules

Completed and verified across the internship period.

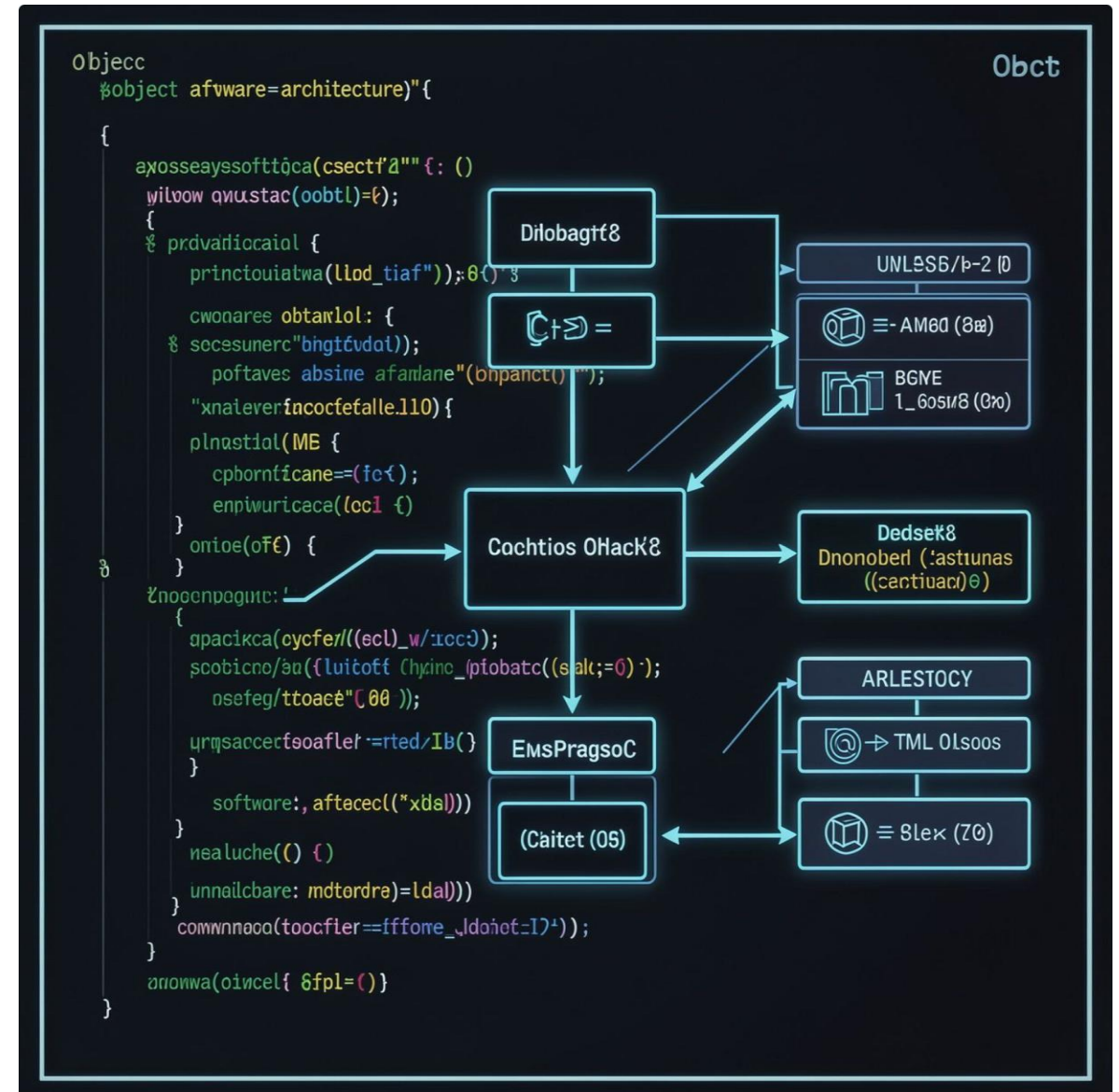
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External DB Dependencies

100% native file persistence through raw C++ streams.

Key Learnings

- **OOP Architecture:** Modular class design, encapsulation, and clear interface separation.
- **File Persistence:** Mastered dynamic stream handling (fstream) for reliable storage.
- **Algorithmic Thinking:** 2D matrix manipulation, loop optimisations, and state validation.





Thank you for reviewing my Virtual Internship Presentation!

Harshit Pandey

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B.Tech CSE (Batch 2026–2027),
Greater Noida

Organisation

Thiranex — Skill Development & Future Tech